

HEIA

Cost analysis

Per-user consumption, transaction volumes, and pricing floors — including corrected Google Maps math.

Selfcare & wellness marketplace · Romania-first

Prepared May 2026 (revision 2)

01 · EXECUTIVE SUMMARY

What we found

Heia is exceptionally cheap to run at every realistic scale. The infrastructure floor — the fixed cost of being online with no users — is **€44 / month** (Vercel Pro + Supabase Pro + domain). Above that, per-user costs are fractions of a euro cent at scale.

The dominant variable cost *at meaningful scale* is Google Maps. With realistic usage assumptions ($\approx$ 2 dynamic map loads per client per month, not the 20 used in the first pass), the Maps line stays inside the \$200 free credit until roughly **30,000 MAU**, and stays under €100/month if static-map caching is implemented for studio cards (the pattern Mero uses on `d3ntpzmu1pnqba.cloudfront.net`).

Whether Heia operates as a SaaS-for-salons (Mero model) or a true marketplace (Heia processes payments and absorbs Stripe), the cost-recovery floor for the platform is small enough that pricing decisions are decoupled from cost: **set price by perceived value, not by infrastructure overhead.**

Phase	Cash floor	Monthly tooling	When
Setup → soft launch	€15 – €500	€44 – €105	Now → +3 months
Romania launch (10k MAU)	€500 – €2,500	€105 – €200	+3 → +9 months
Regional growth (100k MAU)	€2k – €10k	€417 (mitigated)	+12 months on
National scale (1M MAU)	Team build	€2,200 (mitigated)	+24 months on

“Mitigated” means static-map caching on studio cards, Cloudflare R2 for image storage, and Cloudflare CDN in front of Vercel. None are required before 50k MAU.

02 · PER-USER CONSUMPTION MODEL

The 10:1 ratio

Working assumptions: **10 clients per 1 expert**, with experts holding ~10× the stored data (galleries, services, availability vs. a client avatar). Egress flips: clients drive the bandwidth because they browse experts' image-heavy galleries.

Per user / month	Client	Expert	Mix (10:1)
Storage (DB + images)	0.3 MB	3.0 MB	0.55 MB
Egress (bandwidth)	33 MB	5 MB	30 MB
Map loads (Google)	2 – 3	0	~2.2
Emails sent	2.5	1.5	~2.4
Bookings / month	0.30	—	0.27

Reasoning: a client holds an avatar (≈300 KB) and some booking history; an expert holds 6–12 gallery photos + services + availability (≈3 MB). The 10× ratio is structural. Map loads were 20 in the first revision of this document; that estimate was too aggressive and has been corrected to 2–3 based on realistic browsing patterns (see chapter 5).

Volumes at scale

MAU	Clients	Experts	Storage	Egress	Map loads	Emails
1,000	909	91	0.5 GB	30 GB	1.8k	2.4k
10,000	9,091	909	5.5 GB	300 GB	18k	24k
50,000	45,455	4,545	27 GB	1.5 TB	91k	120k
100,000	90,909	9,091	55 GB	3 TB	182k	240k
500,000	454,545	45,455	273 GB	15 TB	909k	1.2M
1,000,000	909,091	90,909	545 GB	30 TB	1.8M	2.4M

03 · MONTHLY COST

Current stack — no mitigations

Vercel + Supabase + Google Maps + Resend + Sentry + Plausible + domain. Stripe is excluded (transactional, modelled separately).

Line	1k	10k	50k	100k	500k	1M
Vercel (Pro + egress)	€19	€25	€42	€112	€670	€1,395
Supabase (Pro + MAU + egress)	€23	€27	€128	€315	€2,557	€5,533
Google Maps (corrected)	€0	€0	€405	€998	€5,732	€11,650
Resend email	€0	€19	€33	€80	€230	€400
Sentry	€0	€24	€74	€74	€230	€230
Plausible analytics	€8	€18	€19	€45	€99	€179
Domain	€2	€2	€2	€2	€2	€2
Total / month	€52	€115	€703	€1,626	€9,520	€19,389

With mitigations

Mitigations stacked: static-map thumbnails for studio cards, Mapbox for the live map, Cloudflare R2 for image storage, Cloudflare CDN in front of Vercel, WorkOS AuthKit instead of Supabase Auth above 100k MAU (free to 1M MAU).

Line	1k	10k	50k	100k	500k	1M
Vercel (behind CF)	€19	€19	€19	€25	€25	€25
Supabase Postgres only	€23	€23	€55	€85	€120	€120
Auth (Supabase / WorkOS)	incl	incl	incl	incl	€0	€0
Cloudflare R2	€0	€0	€0	€1	€4	€8
Maps (static + Mapbox)	€0	€0	€0	€0	€90	€405
Resend email	€0	€19	€33	€80	€230	€400
Sentry	€0	€24	€74	€74	€230	€230
Plausible analytics	€8	€18	€19	€45	€99	€179
Domain	€2	€2	€2	€2	€2	€2
Total / month	€52	€105	€202	€312	€800	€1,369

04 · TRANSACTIONS

Booking and payment volume

Bookings per client per month vary by maturity. The central estimate (0.30) matches the cost model above; the early estimate matches typical pre-PMF marketplace conversion; the mature estimate matches Mero-level engagement.

MAU	Clients	Early (×0.15)	Central (×0.30)	Mature (×0.45)
1,000	909	136 / mo	273 / mo	409 / mo
10,000	9,091	1,364 / mo	2,727 / mo	4,091 / mo
50,000	45,455	6,818 / mo	13,636 / mo	20,455 / mo
100,000	90,909	13,636 / mo	27,273 / mo	40,909 / mo
500,000	454,545	68,182 / mo	136,364 / mo	204,545 / mo
1,000,000	909,091	136,364 / mo	272,727 / mo	409,091 / mo

GMV and Stripe load (central, full-marketplace mode)

Assumed average ticket: **€50**. Stripe fee per transaction: 1.5% + €0.25 ≈ **€1.00 per €50 booking** ≈ **2% of GMV**. Heia absorbs the Stripe cut.

MAU	Bookings / mo	GMV / month	Stripe load / mo
1,000	273	€13,640	€273
10,000	2,727	€136,400	€2,727
50,000	13,636	€681,800	€13,636
100,000	27,273	€1,363,650	€27,273
500,000	136,364	€6,818,200	€136,364
1,000,000	272,727	€13,636,350	€272,727

Stripe load is what Heia pays to the processor. Above this floor, every cent of commission charged to providers is gross margin.

05 · GOOGLE MAPS (CORRECTED)

Where the cost actually comes from

Google Maps Platform charges per API call, with a **\$200 free credit per month** across all Maps APIs combined. Heia uses only the Maps JavaScript API for the live discover map; no Places, Directions, or Geocoding charges.

API	Price	What triggers it
Maps JavaScript — Dynamic	\$7 / 1,000 loads	Each call to new google.maps.Map()
Maps Static API	\$2 / 1,000 loads	Each PNG image of a map
Places Autocomplete	\$17 / 1,000 sessions	Each search-as-you-type session
Places Details	\$17 – \$32 / 1,000	Each metadata fetch
Directions	\$5 / 1,000	Each route
Geocoding	\$5 / 1,000	Each address → lat/lng

The original number was too aggressive

Revision 1 of this document used **20 dynamic map loads per client per month**, which translated to €11,600/month at 100k MAU. That assumption modelled a power user, not a realistic average.

User type	% of base	Map loads / month
Light (open app 1–2×, browse once)	60%	1
Medium (open 5×, browse 2–3×)	30%	3
Heavy (open 15×, browse 5–6×)	10%	7
Weighted average		~2.2

Corrected cost at 100k MAU under different assumptions

Loads / client / month	Total loads	Gross	After \$200 credit	EUR
1 (heavily optimized)	90,909	\$636	\$436	€405
2 (realistic average)	181,818	\$1,273	\$1,073	€998
3 (high engagement)	272,727	\$1,909	\$1,709	€1,589
7 (aggressive)	636,363	\$4,455	\$4,255	€3,957
20 (rev. 1 estimate)	1,818,180	\$12,727	\$12,527	€11,650

What mitigations actually save

Two cheap moves cut Maps cost to near zero at every scale below 1M MAU:

- **Static map thumbnails** for studio cards. Maps Static API is \$2/1k (3.5× cheaper than dynamic) *and* the resulting PNGs can be cached on your own CDN — which makes repeat views free. Mero does exactly this on `d3ntpzmu1pnqba.cloudfront.net`.
- **Only render the live map on /discover**, never on /studio/[id] or in lists. Studio profile pages show a static PNG of the location, not a live embed.

With both, the effective rate drops from ~2.2 loads/client/month to ~0.5. At 100k MAU that's 45,500 loads/month — well inside the \$200 free credit. **Maps cost: €0.**

Plug-and-play formula

$monthly_maps_cost_EUR = \max(0, (MAU \times 0.909 \times loads_per_client \times 0.007) - 200) \times 0.93$

- MAU: total monthly active users
- 0.909: client share (10/11 of MAU)
- loads_per_client: average dynamic map loads per client per month
- \$200: monthly free credit
- 0.93: USD → EUR conversion

06 · PRICING FLOOR

The minimum to keep the lights on

Cost-recovery floor only — pure infrastructure and (where applicable) Stripe fees that Heia absorbs. Whatever margin you set goes on top.

Floor A — Booking-only model (no Stripe, infra only)

MAU	Infra / mo	Floor per expert / mo	Floor per booking
1,000	€52	€0.57	€0.19
10,000	€105	€0.12	€0.04
50,000	€242	€0.053	€0.018
100,000	€417	€0.046	€0.015
500,000	€1,320	€0.029	€0.010
1,000,000	€2,204	€0.024	€0.008

Floor B — Marketplace model (Heia absorbs Stripe)

Stripe = €1 / booking on average $\approx$ 3 bookings / expert / month $\approx$ €3 / expert / month.

MAU	Infra / expert	Stripe / expert	Floor per expert / mo	Floor per booking	Floor as % of GMV
1,000	€0.57	€3.00	€3.57	€1.19	2.4%
10,000	€0.12	€3.00	€3.12	€1.04	2.1%
50,000	€0.05	€3.00	€3.05	€1.02	2.0%
100,000	€0.05	€3.00	€3.05	€1.015	2.0%
500,000	€0.03	€3.00	€3.03	€1.010	2.0%
1,000,000	€0.02	€3.00	€3.02	€1.008	2.0%

Above ~5k MAU the floor essentially flatlines at ~€3/expert/month or ~2% of GMV. Infra is a rounding error from there onward — the floor is dominated by Stripe.

What you can charge above the floor

Example: 10k MAU, marketplace mode. The Stripe + infra floor is €1.04 per €50 booking. The Mero/Booksy market range is 5–15% commission. So if Heia charges:

Headline rate	On €50 booking	Floor	Heia margin / booking
3% of GMV	€1.50	€1.04	€0.46
5% of GMV	€2.50	€1.04	€1.46
10% of GMV	€5.00	€1.04	€3.96
15% of GMV	€7.50	€1.04	€6.46

Or in subscription form at 10k MAU (booking-only, no Stripe):

Headline rate	Stripe	Infra	Heia margin / expert / mo
€5 / expert / mo	€0	€0.12	€4.88
€15 / expert / mo	€0	€0.12	€14.88
€30 / expert / mo (Mero range)	€0	€0.12	€29.88

07 · COMPETITIVE READ

What Mero actually does

Two facts from public data and direct site inspection that shape Heia's positioning.

1. In-app payments are opt-in per salon, not platform-wide

- Mero's primary model is SaaS subscription to the salon (€30–€60/month range).
- In-app card payment via Stripe is an *optional* feature — turned on per salon to reduce no-shows and enable deposits.
- Stripe fee passed through at 1.5% + €1 RON; **no marketplace commission** taken by Mero on top.
- Implication for Heia: differentiation cannot be “we have in-app payments” — Mero already does. Differentiation has to be brand/positioning + payment model (take-rate marketplace vs SaaS subscription).

2. Image architecture: S3 + CloudFront, with pre-sized variants

- Storage: AWS S3 in **eu-central-1 (Frankfurt)**, bucket *novabooker-uploads-eu-central*. (Mero's parent was originally Novabooker.) EU region for GDPR.
- Delivery: AWS CloudFront. ~441 of 461 image references on a typical salon page are CDN-served.
- Images stored as files; database (MongoDB, evidenced by ObjectID-shaped filenames) holds only IDs.
- Variants pre-generated at upload, not on-the-fly: *thumbnail / small / medium / large / croppedLarge*. Trades flexibility for cost — Cloudinary-style transforms would be 10× more expensive.
- Static map thumbnails cached on a separate CloudFront distribution. This is the same Maps mitigation Heia should adopt.

Heia's current stack (Supabase Storage + Supabase CDN) is architecturally equivalent — Supabase Storage is S3-compatible under the hood. The patterns to mirror: pre-sized variants on upload, image references in DB (not bytes), static map caching on a CDN. None of this requires changing stacks.

08 · WATCHPOINTS

Triggers to revisit the stack

Observable signals that mean it is time to migrate one piece. Migrate one piece at a time, never the whole stack at once.

- **Maps spend exceeds €100/month for two consecutive months** → implement static-map caching for studio cards *before* considering Mapbox migration.
- **Supabase DB exceeds 6 GB or query p95 exceeds 200 ms** → upgrade Compute (Small → Medium → Large) before considering a different provider.
- **Studio image storage exceeds 80 GB** → move new uploads to Cloudflare R2 (no egress fees); lazily migrate old ones.
- **Email volume exceeds 30k/month** → upgrade Resend tier or add Postmark for transactional vs marketing split.
- **Vercel bandwidth exceeds 800 GB/month** → put Cloudflare in front of Vercel (free) for image and static asset caching.
- **Supabase MAU exceeds 100k** → evaluate WorkOS AuthKit (free to 1M MAU) for the auth surface, keeping Supabase for Postgres only.

09 · RECOMMENDATION

What to do

Stay on the current stack — Next.js, Vercel, Supabase, Google Maps — through pilot and Romania launch. Total tooling spend stays under €200/month all the way to 10k MAU on any reasonable usage pattern.

Three line items to budget for ahead of launch:

- Vercel Pro and Supabase Pro from launch day onward: ~€42/month combined.
- ~€500 contingency for the first lawyer review of T&Cs and privacy policy, ideally before opening paid bookings.
- €1,500–€3,000 marketing-channel test budget in the first three months of the Romania launch — out of scope here but the actual cost lever.

Re-run this analysis when any watchpoint trigger fires, or every six months, whichever comes first.

This is revision 2 of the cost analysis. Revision 1 used inflated Google Maps assumptions (20 loads/client/month); revision 2 uses the corrected 2–3 loads/client/month based on realistic browsing patterns. The pricing-floor and transaction-volume sections are new.